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Lipid nanoparticle delivery of circle RNA vaccine induces potent immune responses

Ruohan Yang, Lin Jia & Jiuwei Cui✉

Circle RNA (circRNA) is a closed-loop nucleic acid resistant to exonuclease degradation. Their enhanced stability and potential for efficient protein expression through circRNA modification render them promising novel therapeutic agents for cancer treatment. In this study, we systematically investigated the antitumor and anti-inflammatory activities of circRNA vaccines in mice, as well as their effects on activating antigen-specific immune responses, through a meta-analysis. Subsequently, we further validated the immune-activating effects of circRNA using monocytes and T cells isolated from human peripheral blood mononuclear cells. We found that circRNA induces the activation and maturation of dendritic cells, as well as the responses of CD4 + and CD8 + T cells. These findings demonstrate the potential of circRNA to exert significant effects in therapeutic cancer vaccines.

Keywords circRNA, Lipid nanoparticles, Immunotherapy, Malignant tumors, Antigen-presenting cells

Circle RNA (circRNA) constitutes a distinct RNA subtype featuring a covalently closed single-stranded structure[1]. circRNA exhibit non-coding functions such as transcriptional regulation, splicing regulation, signal transduction, and molecular sponge activities. These regulatory roles enable them to promote the onset and progression of various diseases, thereby influencing the intrinsic mechanisms of disease[2–5].

In recent years, the use of in vitro-synthesized RNA to express neoantigens for disease treatment has become a focal point for many researchers[6]. This approach enables the precise and efficient expression of specific proteins within cells and tissues[7]. Currently, the use of mRNA to encode COVID-19 antigens represents one of the most prominent applications of this method. However, mRNA vaccines suffer from poor stability, imposing stringent requirements on transport and storage conditions[8]. Concurrently, mRNA faces the risk of rapid degradation, resulting in short-lived neoantigen expression and necessitating repeated dosing for therapeutic efficacy[9]. Our prior research has demonstrated that, compared to mRNA, circRNA possesses a unique circular structure that confers resistance to RNase R digestion, thereby enhancing its stability and positioning it as a novel vaccine candidate[10]. Studies indicate that circRNA has a half-life approximately 2.5 times longer than mRNA, thereby prolonging the duration of protein expression[11]. As illustrated in Fig. 1, circRNA can mediate ribosomal initiation and protein translation via the internal ribosomal entry site (IRES) element, even in the absence of a free 5′ end, thereby enabling efficient antigen presentation[12]. Moreover, unlike DNA vaccines, circular RNAs carry no risk of host genome recombination[13] and do not require endosomal processing and cleavage, which can reduce antigen presentation efficiency in peptide vaccines[14]. Established research demonstrates that circRNAs can elicit potent T- and B-cell responses, resulting in complete tumor regression in mouse models of melanoma[15]. Wei et al. confirmed that circRNA vaccines provide robust protection against SARS-CoV-2 variants in both mice and rhesus monkeys, exhibiting excellent tolerability with no significant adverse effects[16].

circRNA holds immense potential as a next-generation RNA vaccine platform. Its unique advantages of efficient synthesis and stable high expression have demonstrated promise in vaccines, immunotherapies, and protein replacement therapies, offering innovative and effective treatment prospects for human diseases[17]. Concurrently, integrating lipid nanoparticles (LNPs) with RNA vaccines significantly enhances their therapeutic applications[18]. LNP efficiently encapsulates RNA, enabling large-scale and cost-effective production.

Therefore, building upon these findings, this study investigates the extent to which LNP-encapsulated circRNA induces immune responses in mice, thereby exerting antitumor and anti-inflammatory effects. This research provides valuable insights and perspectives for future investigations.

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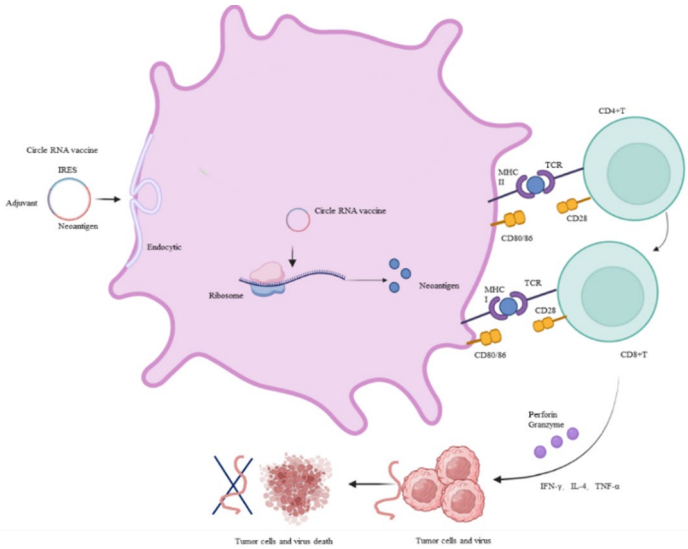


Fig. 1. Mechanism of action diagram for circRNA vaccines.

Materials and methods

Search strategy

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[19]. Internet searches were conducted across the PubMed, Cochrane Library, EMBASE, Springer, and ScienceDirect, and OVID databases to retrieve relevant publications from 2020 to the present (October 2025). The following search terms were used to identify articles: 'circle RNA', 'circRNA", circular RNA', and 'vaccine'. (PROSPERO ID: CRD420251242287).

Inclusion criteria

Inclusion criteria for meta-analysis studies: (1) circRNA therapy for tumors or infectious diseases delivered via LNP; (2) Detailed documentation of intervention measures, experimental groupings, and dosage administration; (3) Inclusion of at least one of the following metrics: tumor volume, mouse body weight, and immunological parameters; (4) Full-text articles available.

Exclusion criteria

Exclusion criteria were as follows: (1) non-mouse animal models or in vitro studies; (2) Data failing to demonstrate tumor changes in mice before and after treatment; (3) Reviews, communications, case reports, presentations, letters, expert opinions, and editorials; (4) Relevant data unavailable for extraction; (5) Failure to meet inclusion criteria.

Data extraction and quality assessment

The current studies primarily comprised preclinical research using animal models. We analyzed data on whether mice with tumors or infectious diseases benefit from treatment with a circRNA vaccine. Extracted content included: (a) basic study details, including first author and publication year; (b) study characteristics, including tumor volume, weight, inhibition rate, mouse body weight, and adverse reactions; (c) elements relevant to risk of bias assessment. Using tools developed by the SYRCLE[20], the authors assessed the methodological quality and risk of bias in the included trials across domains: selection, performance, detection, attrition, reporting, and other biases.

Plasmid construction

The vector was purchased from MIAOLING Biotech (IRES-Circ-GFP-FLAG). Immunofluorescence detection was subsequently performed using an anti-Flag antibody (Sigma-Aldrich, F1804). circRNA was synthesized using the HiScribe T7 High Yield RNA Synthesis Kit (NEB E2040S). IVT templates were amplified via PCR before RNA synthesis (Q5 Hot Start High Fidelity 2x Master Mix). Each 20 µL IVT reaction utilized 1 µg of circRNA PCR template. Reactions proceeded at 37 °C for 2 h. Subsequently, IVT templates were degraded by incubating with 2 µL DNase I (NEB M0303S) at 37 °C for 20 min. Digestion was performed using 1 U RNase R per µg RNA at 37 °C for 30 min, followed by RNase R inactivation at 70 °C for 10 min. Circularization efficiency was assessed via RT-qPCR (forward primer: TCTGAACCACACAAATGGTC; reverse primer: ATGGTGGGG TACCCATCCGCT).

Cell lines

Healthy human peripheral blood mononuclear cells (PBMCs) were commercially purchased from Beijing VITONLIVIA (D176368). Cell line origin: human. Gender: male. Source tissue: peripheral blood. The cell lines used were not misidentified, and there was no contamination during the experiment.

CD14 + magnetic beads (MicroBeads, Miltenyi, 130-050-201) were added at a ratio of 20 µl per 10^7 PBMCs and incubated at 4 °C in the dark for 15 min. Following incubation, peripheral blood mononuclear cells were isolated using LS Columns (MASC, 130-042-401). Cells were cultured in 10% 1640 medium supplemented with cytokines IL-4 (50 ng/ml) and GM-CSF (20 ng/ml) to induce dendritic cell differentiation.

For in vitro T-cell activation assays, healthy PBMCs were also commercially purchased from Beijing VITONLIVIA(D176368). CD3 + magnetic beads (MicroBeads, Miltenyi, 130-097-043). For every 10^7 PBMCs, add 10 µl $CD14^+$ magnetic beads and incubate at 4 °C in the dark for 15 min. Subsequently, isolate $CD3^+$ T cells using an LS Column and culture them in 1640 medium, promptly adding activated dendritic cells to maintain activity. The culture medium contains 10% foetal bovine serum (FBS; Gibco), 2% penicillin, 1% streptomycin, and 1% levofloxacin, and cells are grown in a humidified environment at 37 °C with 5% CO_2. All cell lines are monitored for mycoplasma infection using the Impact III PCR method.

Flow cytometry analysis

The following antibodies target surface markers: Human CD4 (FITC, BioLegend, 300505, 1:100); Human CD8 (APCy7, BioLegend, 300908, 1:100); human IFN-γ (BD, PE, 344117,1:100; human CD80 (APC, BioLegend, 305219, 1:100); human CD86 (FITC, BioLegend, 344704, 1:100).

Immunofluorescence

In immunofluorescence experiments, transfected dendritic cells were centrifuged and smeared onto microscope slides. Sections were incubated overnight with monoclonal anti-FLAG M2 mouse antibody (Sigma, F1804, 1:500 dilution), followed by incubation with Alexa Fluor 647-labelled secondary antibody. DAPI was used for nuclear counterstaining. Images were acquired using a laser scanning confocal microscope equipped with NIS-Elements Viewer imaging software.

RT-qpcr

RT-qPCR analysis was performed in triplicate using Brilliant II SYBR Green qRT-PCR Master Mix (Agilent, 600825) and a LightCycler 480 (Roche). The relative RNA level was calculated by the ddCt method compared to the B-Actin control.

Primer sequences:
IL-6 Forward: AGGAGACTTGCCTGGTGAAA
Reverse: CAGGGGTGGTTATTGCATCT.
IL-Iβ Forward: TGACGGACCCCAAAAGATGA.
Reverse: AAAGACACAGGTAGCTGCCA.
TNF-α Forward: GAGGCCAAGCCCTGGTATG.
Reverse: CGGGCCGATTGATCTCAGC.

Preparation and purification of in vitro transcribed circRNA

To eliminate double-stranded RNA (dsRNA) contaminants, the crude transcription reaction was directly treated with RNase III at 37 °C for 30 min to specifically degrade dsRNA. The reaction was then purified using a silica-membrane-based RNA cleanup column to remove proteins, salts, and unincorporated nucleotides. And the absence of dsRNA contamination was verified using a dsRNA-specific fluorometric quantification kit and ELISA with a four-parameter logistic (4PL) standard curve. (Yesaen, 36717ES96).

ELISpot

As mentioned above, ELISpot T cells were collected from peripheral blood and then restimulated with dendritic cells (DCs) transfected with circleRNA for 24 h. IFN-γ ELISpot assays were performed according to the manufacturer's instructions (Dakewe, 2110005, Human) to detect IFN-γ-secreting T cells, and the colored spots were counted.

Statistical analysis

All data were analyzed using Review Manager 5.4 software. Forest plots displayed pooled risk ratios with their respective 95% confidence intervals. Statistical heterogeneity was assessed using the inconsistency index (I^2). Data exhibiting substantial heterogeneity were analyzed using random-effects models, whilst those with minimal heterogeneity were evaluated using fixed-effects models.

Prism software version 9.2.0 was used to perform statistical analyses. One-way and two-way ANOVA tests and an unpaired two-sided Student's t test were used to compare more than two groups, and differences between groups with P-values below 0.05 were considered significant.

Results

Systematic screening

The study screening process is illustrated in Fig. 2. A total of 427 articles met the inclusion criteria. Among these, duplicate and irrelevant articles (e.g., studies with inconsistent subjects) were excluded following an abstract review. Subsequently, articles with inconsistent methodologies (e.g., solely in vitro cellular experiments or the absence of tumor-related endpoints) were excluded after full-text review. Ultimately, ten tumors[15,21–29] and eleven infectious[16,30–39] disease model preclinical studies were selected for meta-analysis.

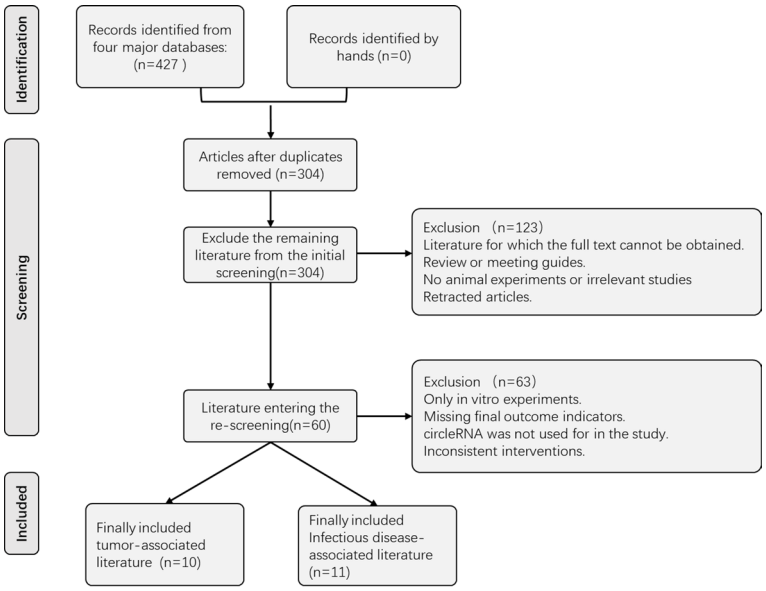


Fig. 2. Literature screening process.

Basic demographic characteristics of included studies

Data extracted from the included studies are presented in Table 1. All selected studies were randomized controlled trials (RCTs) utilizing mice as experimental subjects. Overall, the animal strains employed were predominantly BLAB/C or C57BL/6, with administration primarily via intramuscular or subcutaneous injection. Notably, within infectious disease mouse models, three instances involved intranasal administration, and one case employed tracheal instillation. Among animal tumor models, ovalbumin served as the neoantigen in four instances, representing its most prevalent application.

Risk of bias assessment

A total of 10 studies reported the baseline characteristics of experimental mice, with no significant bias identified. The summary of preliminary risk assessment results for selected articles is presented in Table 2.

Body weight changes in all mice before and after treatment

As shown in Fig. 3, we compared the body weight of mice before and after treatment according to different administration routes. Subgroup analyses were performed for subcutaneous injection, intramuscular injection, intranasal injection, intravenous injection, and other administration routes (including intraperitoneal injection, intramuscular injection, and intratracheal instillation). Compared with the control group, there was no statistically significant difference in body weight in the vaccine group across different administration routes (overall standardized mean difference (SMD) = -0.05, 95% CI = -0.42 to 0.32, $P = 0.80$). Subgroup analyses showed that the pooled effect sizes of different administration routes did not reach statistical significance, and the overall between-study heterogeneity was low ($I^2 = 0\%$).

Changes in T-cell proportion before and after treatment in all mice

The results demonstrated that the vaccine group exhibited a significantly increased proportion of activated T cells compared with the control group (overall SMD = 4.14, 95% CI = 2.67 to 5.61, $P < 0.00001$). Subgroup analyses indicated that the vaccine conferred significantly superior effects relative to the control via subcutaneous, intramuscular, and intravenous administration (all $P < 0.05$), whereas no statistically significant effect was observed in the subgroup of other administration routes (Fig. 4).

Assessment of tumor-related parameters in mice

Subsequently, we conducted separate efficacy evaluations on mice within the tumor model, including measurements of tumor volume changes. We also documented alterations in immune cells playing pivotal roles in tumor treatment, such as macrophages and dendritic cells (DCs).

Tumor volume changes

Figure 5 summarizes tumor volume alterations in mice before and after circRNA treatment. Compared with the control group, the vaccine immunization group exhibited a significantly reduced tumor volume in mice (overall SMD = -5.36, 95% CI: -7.52 to -3.20, $P < 0.00001$), although high between-study heterogeneity was observed ($I^2 = 70\%$, $P = 0.002$). Subgroup analyses stratified by administration route showed statistically significant effects in the subcutaneous injection subgroup (SMD = -6.22, 95% CI: -11.39 to -1.04, $P = 0.02$) and intravenous injection subgroup (SMD = -7.90, 95% CI: -10.94 to -4.86, $P < 0.00001$). The intramuscular injection subgroup showed a borderline statistical trend (SMD = -4.77, 95% CI: -9.70 to 0.15, $P = 0.06$), whereas no statistical significance was

Author	Publication date	Neoantigen	Animal strains	Cell line	Method of administration	Liposome composition
Tumor						
Alshehry et al.[21]	2025	Ovalbumin, human papillomavirus (HPV)-16 E7 antigen	C57BL/6	NA	Subcutaneous injection	SM-102, cholesterol, DOPE, and DMG-PEG2000
Amaya et al.[15]	2023	ovalbumin	C57BL/6	B16-F10-OVA	Intraperitoneal injection	Oleyl- (O) and nonenyl-substituted (N) carbonate subunits followed by a block of on average nine α-amino ester subunits (A)
Cai et al.[27]	2024	FAPα and survivin	C57BL/6	Panc02	Intramuscular injection	NA
Huang et al.[25]	2023	Personalized Breast Cancer Vaccine circFAM53B (192–200) (ALFRLTNRA) or circFAM53B (210–218) (RTAHYGTGR)	NOD/SCID	B16F10, 4T1	Intravenous injection	NA
Jiang et al.[29]	2025	GPC3	B6-hHLA-A2.1/hB2M	hepa1-6 cells	Subcutaneous injection	Cholesterol, DSPC, DMG-PEG2000, SM102
Li et al.[24]	2022	Ovalbumin	C57BL/6	B16	Intramuscular injection	PAMAM dendrimer G0 was mixed with 1,2-epoxytetradecane
Li et al.[28]	2025	Ovalbumin	C57BL/6	B16-OVA	Intranasal	SM102, DSPC, DMG-PEG, and cholesterol.
Wang et al.[23]	2024	PTPN2	C57BL/6	Hepa1-6	Intravenous injection	commercially available lipids (in vivo-jetRNA).
Wang et al.[22]	2025	Chimeric antigen receptor	BALB/c	CT26, 4T1, MC38-HER2	Intravenous injection	Lipids including 18PA, DOPE, Cholesterol, DMG-PEG2000, and C12-200
Zhang et al.[26]	2025	Ovalbumin	C57BL/6	EG7.OVA, MC38	Subcutaneous injection	DOTAP, cholesterol, and DSPE-PEG2000
Author	**Publication date**	**Neoantigen**	**Animal strains**	**Virus types**	**Method of administration**	**Liposome composition**
Infectious disease						
Huang et al.[37]	2024	Delta receptor binding domain	BALB/c	SARS-CoV-2	Intramuscular injection	AX4, DSPC, DMG-PEG2000
Liu et al.[31]	2024	Dimeric EDIII	C57BL/6	Zika virus	Subcutaneous injection and intramuscularly	SINOPEG, SM102/ DSPC/cholesterol/DMG-PEG 2000
Qu et al.[16]	2022	Trimeric RBD of the spike protein	BALB/c	SARS-CoV-2	Intramuscularly	Commercial LNP
Swingle et al.[34]	2025	Anti-S receptor binding domain (RBD)	C57BL/6	SARS- CoV- 2	Intramuscular injection	12D6.2
Wan et al.[38]	2024	Influenza virus haemagglutinin	BALB/c	influenza and SARS- CoV- 2	Intramuscular injection	SM102, DSPE-PEG - Maleimide
Wan et al.[39]	2024	SAD-L16 glycoprotein (G), SARS-CoV-2 receptor-binding domain (RBD)	C57BL/6	Rabies virus strain CVS-24	Intramuscular injection	SM102, DSPC, cholesterol
Wu et al.[32]	2025	Chimeric immunogen of B6 and M1	BALB/c	MPXV	Intranasally	An ethanol phase containing ionizable lipid, 1,2-distearoyl-sn-glycerol-3-phosphocholine, cholesterol, and lipid-polyethylene glycol
Yue et al.[30]	2024	Neuraminidas	NA	H1N1, H3N2, and influenza B viruses	Intramuscularly	SM102, 1,2-Distearoyl-sn-glycero-3-phosphorylcholine (DSPC), cholesterol, and DMG-PEG2000
Zhang et al.[35]	2025	S1 region of the S protein	BALB/c	Porcine epidemic diarrhea virus and transmissible gastroenteritis virus	Intranasally	NA
Zhang et al.[36]	2025	M2e	BALB/c	Influenza A virus	Intratracheal instillation	Ionizable lipids, 1,2-Distearoyl-sn-glycero-3-phosphorylcholine (DSPC), 1,2- dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (DMG- PEG2000)
Zhou et al.[33]	2024	A29L, A35R, B6R, and M1R, which are located on the EEVs or IMVs of MPXV	BALB/c	Vaccinia virus	Intranasally	ALC-0315, DSPC, cholesterol, ALC 0519

Table 1. Basic characteristics of the included literature.

Items										
	1	2	3	4	5	6	7	8	9	10
Study-tumor										
Alshehry et al. (2025)	-	-	-	-	-	-	-	+	+	+
Amaya et al.(2023)	-	?	-	-	-	-	-	+	+	+
Cai et al. (2024)	-	-	-	-	-	-	-	+	+	+
Huang et al. (2023)	-	?	-	-	-	-	-	+	+	+
Jiang et al. (2025)	?	-	-	-	-	-	-	+	+	+
Li et al.(2022)	?	+	-	+	-	-	-	?	+	+
Li et al.(2025)	-	+	?	-	-	-	-	+	+	+
Wang et al. (2023)	-	+	?	-	-	-	-	?	+	+
Wang et al. (2025)	-	?	-	+	-	-	-	+	+	+
Zhang et al.(2024)	+	-	+	+	-	-	-	+	+	+
Study-infectious disease										
Huang et al.(2024)	-	?	-	-	-	+	-	+	+	+
Liu et al. (2024)	?	-	-	-	+	-	-	+	+	+
Qu et al. (2022)	-	?	-	+	-	-	-	+	+	+
Swingle et al. (2025)	-	-	-	+	-	-	-	+	+	+
Wan et al. (2024)	?	-	-	-	-	-	-	+	+	+
Wan et al. (2024)	-	-	?	-	-	+	-	+	+	+
Wu et al. (2025)	+	-	-	+	-	+	-	+	+	+
Yue et al. (2024)	?	+	+	-	+	-	-	+	+	+
Zhang et al.(2025)	+	+	-	+	-	+	-	+	+	+
Zhang et al.(2025)	+	-	-	-	+	-	-	+	+	+
Zhou et al. (2024)	+	-	?	-	+	+	-	+	+	+

Table 2. Evaluation of literature quality according to SYRCLE. Risk of bias with the SYRCLE tool. "+": low risk; "-": high risk; "?": unclear risk. 1(Selection bias): Was the allocation sequence adequately generated and applied? 2(Selection bias): Were the groups similar at baseline, or were they adjusted for confounders in the analysis? 3(Selection bias): Was the allocation adequately concealed? 4(Performance bias): Were the animals randomly housed during the experiment? 5(Performance bias): Were the caregivers and/or investigators blinded from knowledge of which intervention each animal received during the experiment? 6(Detection bias): Were animals randomly selected for outcome assessment? 7(Detection bias): Was the outcome assessed or blinded? 8(Attrition bias): Were incomplete outcome data adequately addressed? 9(Reporting bias): Are reports of the study free of selective outcome reporting? 10(Other): Was the study free of other problems that could result in a high risk of bias?

reached in the other administration routes subgroup (SMD = -1.67, 95% CI: -3.93 to 0.60, $P = 0.15$). Notably, significant differences in effect size were detected among different administration route subgroups ($P = 0.01$), suggesting that administration route represents an important source of heterogeneity.

Immunocellular changes

As shown in Fig. 6, compared with the control group, intervention in the vaccine group significantly increased the proportion of macrophages overall SMD = 4.88 (95% CI: 1.65 to 8.11, $P = 0.003$), with high between-study heterogeneity ($I^2 = 79\%$, $P = 0.0008$). Subgroup analyses stratified by administration route revealed a statistically significant effect in the subcutaneous injection subgroup (SMD = 5.35, 95% CI: 1.63 to 9.08, $P = 0.005$) with moderate-to-high heterogeneity ($I^2 = 63\%$). The effect in the intravenous injection subgroup did not reach statistical significance (SMD = 4.37, 95% CI: -3.65 to 12.38, $P = 0.29$) and showed extremely high heterogeneity ($I^2 = 89\%$). The other administration routes subgroup exhibited borderline statistical significance (SMD = 6.23, 95% CI: 0.00 to 12.45, $P = 0.05$). The effect in the intramuscular injection subgroup could not be estimated due to missing data.

Regarding dendritic cell expression levels, the vaccine group significantly elevated dendritic cell levels overall, thereby enhancing antigen-presenting capacity (overall SMD = 8.02, 95% CI: 4.34 to 11.69, $P < 0.0001$), with high between-study heterogeneity ($I^2 = 84\%$, $P < 0.0001$). All administration routes except the other routes subgroup showed statistical significance: subcutaneous injection (SMD = 9.91, 95% CI: 1.00 to 19.81, $P = 0.05$) with extremely high heterogeneity ($I^2 = 90\%$); intramuscular injection (SMD = 13.94, 95% CI: 8.70 to 19.17, $P < 0.00001$); and intravenous injection (SMD = 7.59, 95% CI: 4.65 to 10.53, $P < 0.00001$) with no heterogeneity ($I^2 = 0\%$, Fig. 7). Significant differences in effect size were observed among subgroups ($P = 0.0004$, $I^2 = 83.6\%$), indicating that administration route is an important source of between-study heterogeneity.

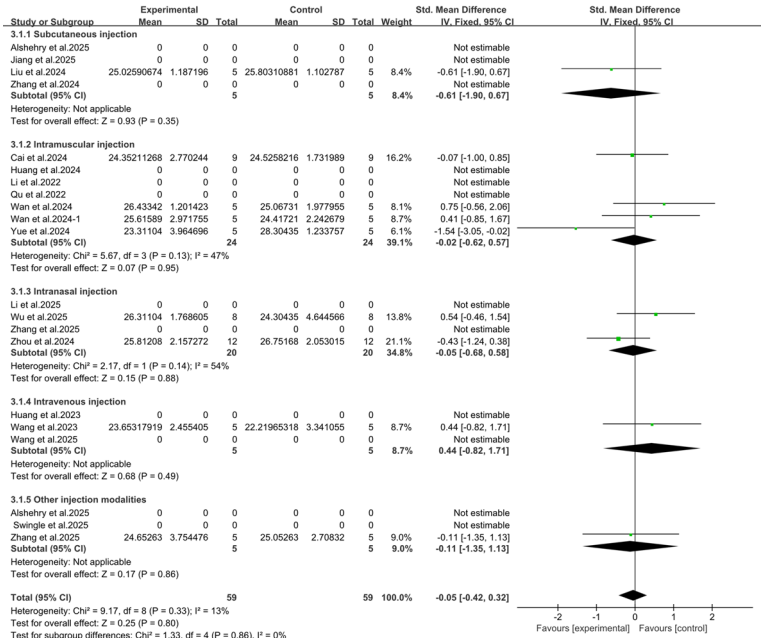


Fig. 3. Subgroup analysis of mouse body weight according to different administration routes. Forest plot showing the standardized mean difference of the mouse body weight between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are presented with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

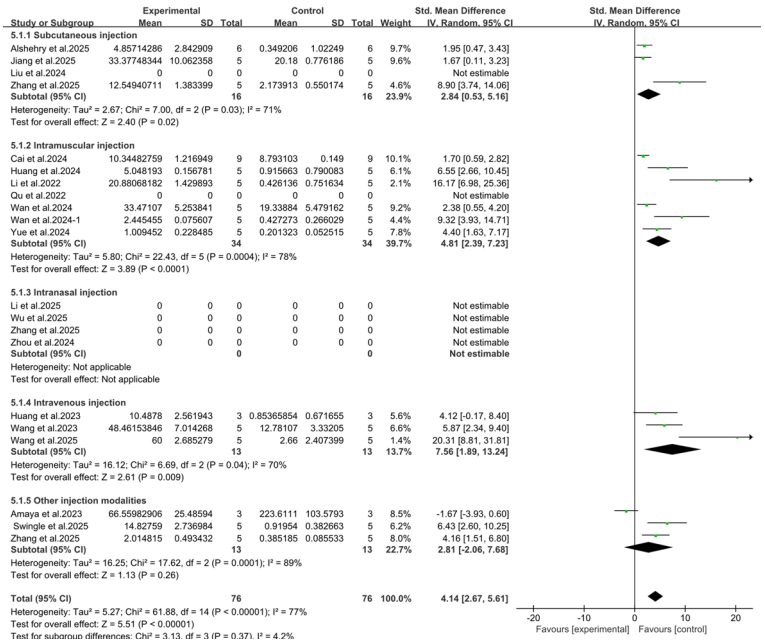


Fig. 4. Subgroup analysis of T-cell proportions according to different administration routes. Forest plot showing the standardized mean difference of the proportion of activated T cells between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are presented with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

Detection of infectious disease-related indicators

Concurrently, we placed significant emphasis on evaluating the therapeutic efficacy of our compounds in infectious disease mouse models. Thus, serum antibody titers and viral load were selected to assess treatment outcomes, while cytokines IFN-γ and IL-4 were chosen to evaluate the effects on the immune system response.

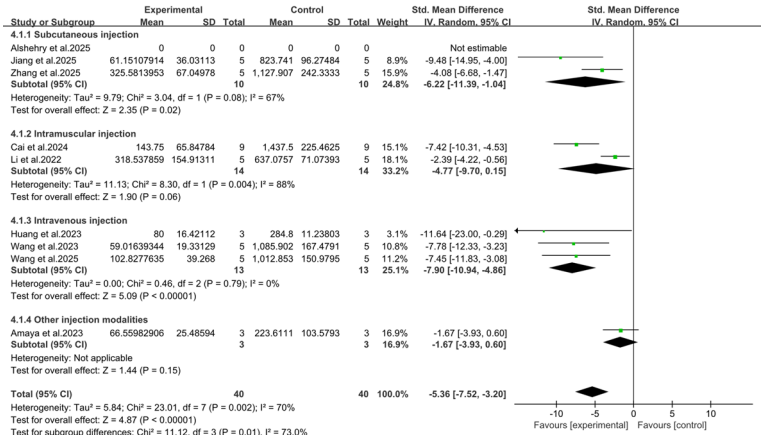


Fig. 5. Subgroup analysis of Tumor Volume Changes according to different administration routes. Forest plot showing the standardized mean difference of the tumor volume changes between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are presented with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

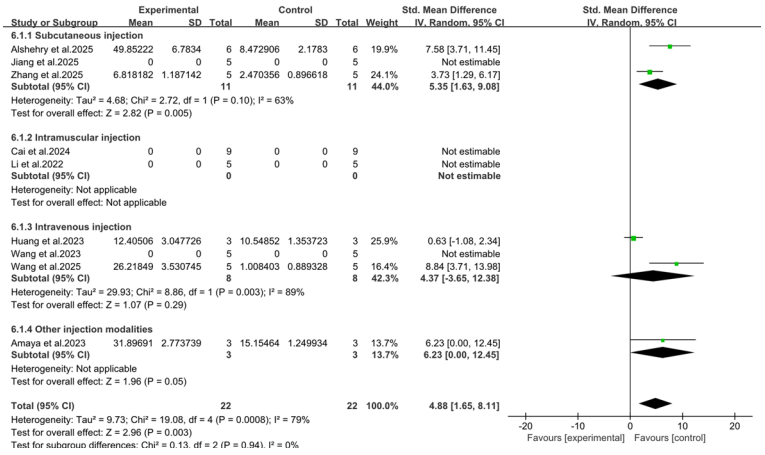


Fig. 6. Subgroup analysis of macrophage changes according to different administration routes. Forest plot of the standardized mean difference in macrophage proportion between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are shown with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

Changes in serum antibody titres

As illustrated in Fig. 8, serum antibody titres in mice with infectious diseases showed a marked increase following circular RNA treatment. Compared with the control group, intervention in the experimental group significantly increased serum antibody titres (overall SMD = 5.85, 95% CI: 3.69 to 8.02, $P < 0.0001$), with substantial between-study heterogeneity (I^2 = 79%, $P < 0.0001$). Subgroup analyses stratified by administration route demonstrated significant effects across all routes: subcutaneous injection (SMD = 5.93, 95% CI: 2.82 to 9.04, $P = 0.0002$); intramuscular injection (SMD = 6.08, 95% CI: 1.86 to 10.29, $P = 0.005$); intranasal injection (SMD = 8.10, 95% CI: 5.10 to 11.09, $P < 0.00001$); and other administration routes (SMD = 2.86, 95% CI: 0.24 to 5.79, $P = 0.03$). Significant differences in effect size were detected among subgroups ($P = 0.004$, I^2 = 63.2%), suggesting that administration route serves as an important source of between-study heterogeneity.

As shown in Fig. 9, intervention in the vaccine group significantly reduced pulmonary viral load overall (overall SMD = -9.50, 95% CI: -14.70 to -4.30, $P = 0.0003$), accompanied by high between-study heterogeneity (I^2 = 81%, $P = 0.001$). Subgroup analyses stratified by administration route revealed significant effects for all included routes: subcutaneous injection (SMD = -5.13, 95% CI: -7.87 to -2.38, $P = 0.0003$); intramuscular injection (SMD = -9.25, 95% CI: -17.30 to -1.19, $P = 0.02$); and intranasal injection (SMD = -15.14, 95% CI: -19.89 to -10.38, $P < 0.00001$). The effect for the other administration routes subgroup could not be estimated due to insufficient data. Significant differences in effect size were observed among subgroups ($P = 0.002$, I^2 = 84.5%), indicating that administration route represents a major source of between-study heterogeneity.

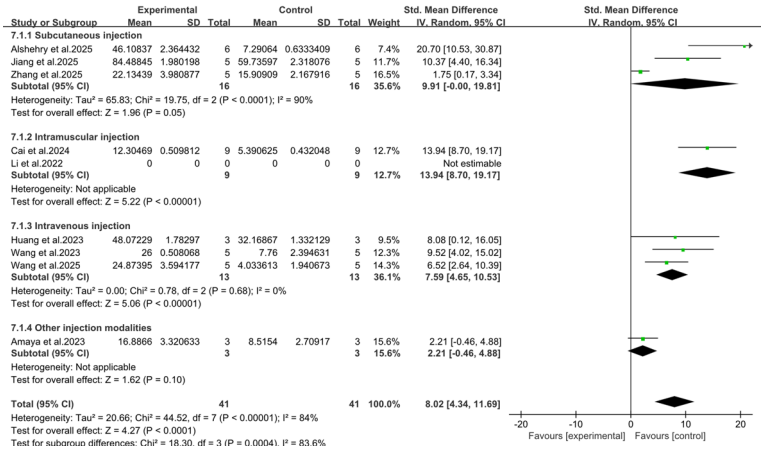


Fig. 7. Subgroup analysis of dendritic cell Changes according to different administration routes. Forest plot of the standardized mean difference (SMD) in dendritic cell levels between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are shown with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

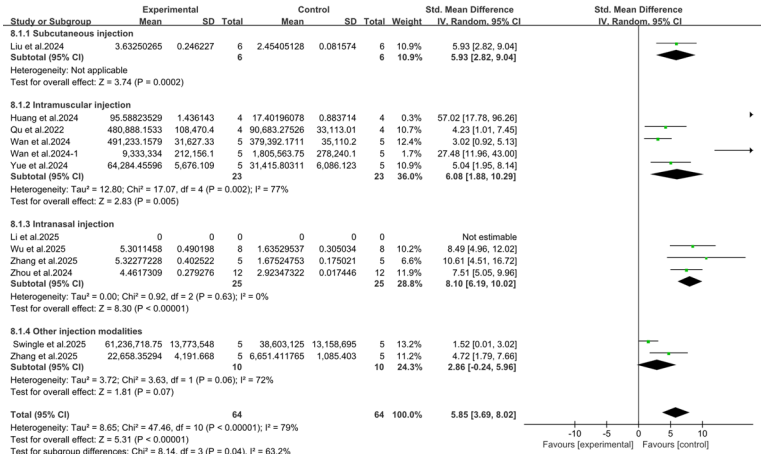


Fig. 8. Subgroup analysis of serum antibody titres according to different administration routes. Forest plot of the standardized mean difference in serum antibody titres between the experimental and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are presented with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

Serum cytokine changes

We also examined serum cytokine alterations. As shown in Fig. 10, the vaccine group exhibited significantly elevated IFN-γ levels overall SMD = 3.11, 95% CI: 1.54 to 4.69, $P = 0.0001$), with high between-study heterogeneity (I^2 = 80%, $P < 0.0001$). Subgroup analyses stratified by administration route demonstrated significant effects across all routes: subcutaneous injection (SMD = 1.69, 95% CI: 0.29 to 3.09, $P = 0.02$); intramuscular injection (SMD = 2.59, 95% CI: 1.12 to 4.06, $P = 0.0005$); intranasal injection (SMD = 8.40, 95% CI: 0.50 to 16.30, $P = 0.01$); and other administration routes (SMD = 2.49, 95% CI: 0.02 to 4.97, $P = 0.04$).

Subsequently, we found that IL-4 levels were also significantly increased in the vaccine group (overall SMD = 2.98, 95% CI: 1.54 to 4.43, $P < 0.0001$), with moderate-to-high between-study heterogeneity (I^2 = 65%, $P = 0.009$). Subgroup analyses stratified by administration route showed significant effects: intramuscular injection (SMD = 2.29, 95% CI: 1.05 to 3.53, $P = 0.0003$); intranasal injection (SMD = 9.10, 95% CI: 6.57 to 24.76, $P = 0.26$); and other administration routes (SMD = 8.83, 95% CI: 3.71 to 13.96, $P = 0.0007$). The effect for the subcutaneous injection subgroup could not be estimated due to unavailability of data. Significant differences in effect size were detected among subgroups ($P = 0.04$, I^2 = 69.5%), suggesting that administration route serves as an important source of between-study heterogeneity (Fig. 11).

In vitro experiments validate circRNA's capacity to induce immune responses

To further validate that circRNA can induce robust immune responses, we designed a circRNA vaccine capable of expressing a FLAG tag and encoding GFP. Following circularization and digestion, circRNA constituted over

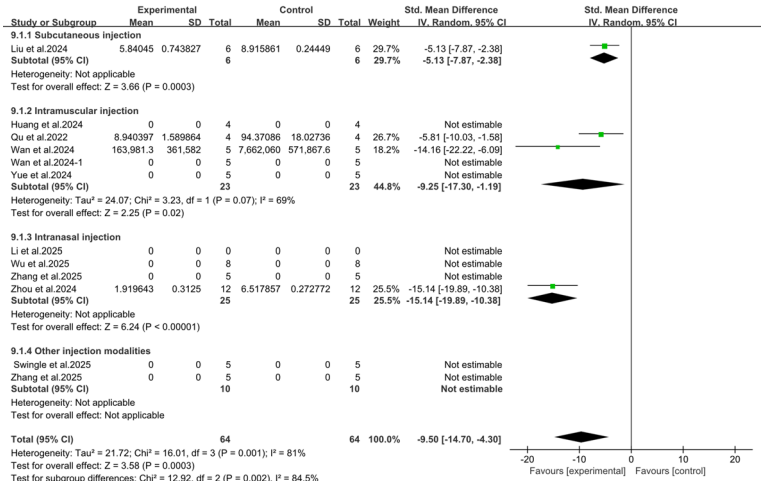


Fig. 9. Subgroup analysis of pulmonary viral load according to different administration routes. Forest plot of the standardized mean difference in pulmonary viral load between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are shown with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

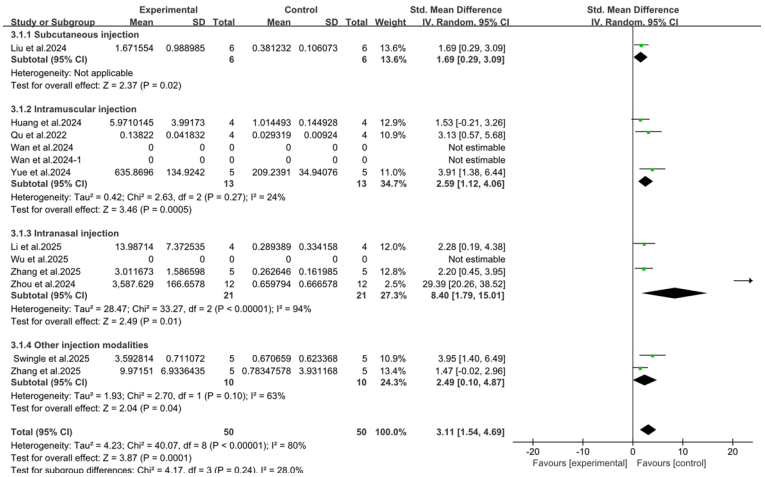


Fig. 10. Subgroup analysis of serum IFN-γ levels according to different administration routes. Forest plot of the standardized mean difference in serum IFN-γ levels between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are presented with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

80% of the product (Fig. 12A). To minimize the immune-stimulatory effect of dsRNA contaminants generated during in vitro transcription (IVT), we performed an additional digestion with RNase III, followed by silica column purification. The dsRNA content in the circRNA preparation was then determined by ELISA. The results showed that the dsRNA level was significantly reduced after digestion (Fig. 12B). Following transfection, dendritic cells exhibited significant upregulation of the cytokines IL-6, IL-1β, and TNF-α (Fig. 12C). As anticipated, immunofluorescence revealed that circRNA-transfected dendritic cells successfully expressed Flag (Fig. 12D).

Following circRNA delivery to dendritic cells via LNP the proportion of CD80.86-positive dendritic cells markedly increased (Fig. 13C). Subsequently, we co-cultured activated mature DCs with T cells for 24 h, then conduct flow cytometric test IFN-γ levels in $CD8^+$ T and $CD4^+$ T cells. Results demonstrated activation of both effector CD8 + and CD4 + T cells (Fig. 13A, B).

To investigate the antigen-specific T cells produced by circRNA, DCs were transfected with circRNA or LNP-circRNA, or without transfected, and then co-cultured with T cells. The number of IFN-γ-secreting T cells was determined by ELISpot assay (Fig. 13D). LNP-circRNA DCs transfected with circRNA significantly increased the number of IFN-γ-positive spot-forming cells (SFCs) in co-cultured T cells ($P < 0.0001$). These results demonstrated that LNP-circRNA has a potent capacity to trigger antigen-specific T cell immune responses in vitro.

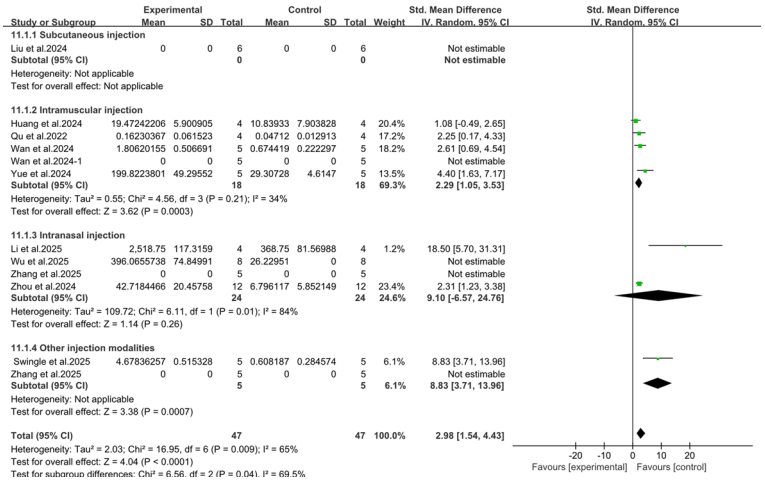


Fig. 11. Subgroup analysis of serum IL-4 levels according to different administration routes. Forest plot of the standardized mean difference in serum IL-4 levels between the vaccine and control groups. Subgroup analyses were performed based on administration routes. The overall effect and subgroup effects are shown with 95% confidence intervals (CIs). $P < 0.05$ was considered statistically significant.

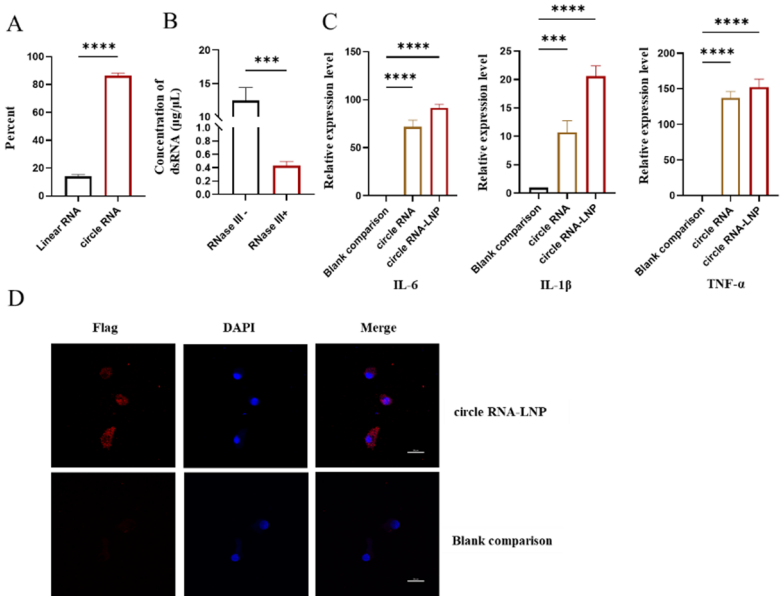


Fig. 12. (A) Proportion of circular and linear RNA following digestion of in vitro transcribed circRNA, data were presented as mean ± SD, an unpaired two-sided Student's t test was performed for comparison; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$; ns, not significant. (B): dsRNA was quantified by ELISA with a four-parameter logistic (4PL) standard curve. Data were presented as mean ± SD. An unpaired two-sided Student's t test was performed for comparison; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$; ns, not significant. (C)Transfected dendritic cells were analyzed for cytokine expression of IL-6, IL-1β, and TNF-α. Repeated-measures ANOVA was applied in the figure. $^{**}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. (D): Transfection of dendritic cells with circRNA enables expression of neoantigens.

Discussion

circRNA is a covalently closed circular RNA molecule that has garnered significant attention due to its exceptional durability and stability, offering promise as a stable platform for gene expression delivery[40,41]. In this study, we first performed a systematic meta-analysis to investigate the therapeutic efficacy of lipid nanoparticle (LNP)-delivered circRNA in mouse tumor models. Consistent and marked tumor regression was observed across all included mouse models, and notably, no significant alterations in body weight were detected throughout the treatment period. This finding not only confirms the potent antitumor activity of circRNA vaccines but also indicates a favorable safety profile, supporting LNP-circRNA as a safe and effective therapeutic strategy for tumor intervention. Accumulating evidence has established that the efficacy of therapeutic tumor vaccines

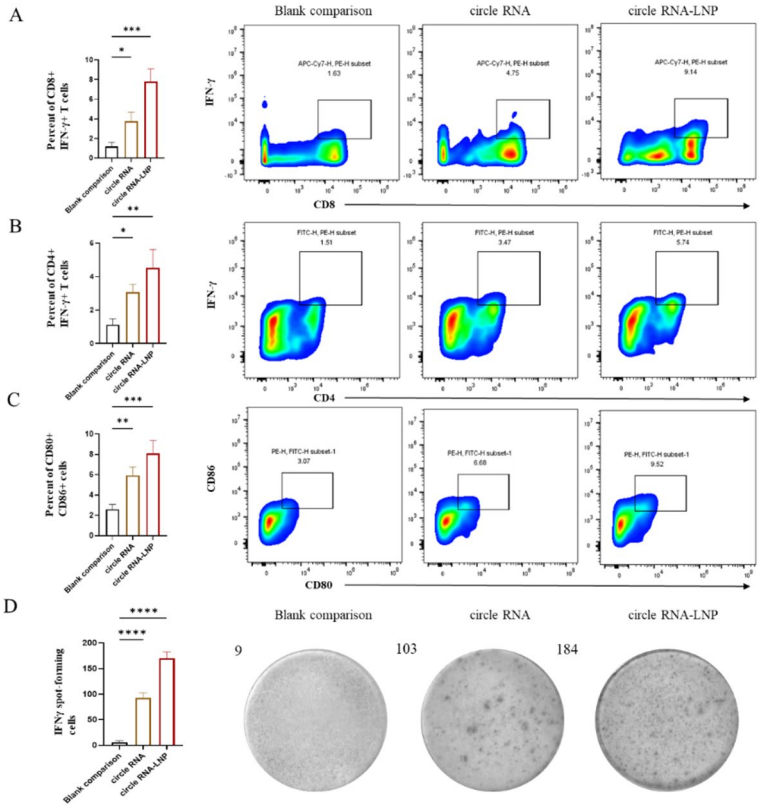


Fig. 13. (A, B) IFN-γ levels in $CD8^+$ T and $CD4^+$ T cells were detected by flow cytometry after 24 h of co-culture between circRNA-transfected DCs and T cells. (C): DCs were transfected with circRNA or LNP-circRNA, or with no transfection. The expression levels of CD80 and CD86 in DCs were analyzed by flow cytometry after 24-hour incubation. (D): DCs were transfected with circRNA or LNP-circRNA, or with no transfection, then co-cultured with T cells. IFN-γ ELISpot assay was performed to detect the number of spot-forming cells (SFCs). Repeated-measures one-way ANOVA was applied in the figure. Data are presented as mean ± SD. $^{**}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$.

is predominantly dependent on efficient antigen presentation and the subsequent activation of dendritic cells (DCs)—the key professional antigen-presenting cells in the immune system[42]. Consequently, we analyzed immune cell states to elucidate the mechanism by which circRNA vaccines exert their effects. Results revealed marked proliferation and activation of T cells, DCs, and other immune cells following vaccination, suggesting that circRNA exerts potent antitumor effects by modulating the innate immune system. These results collectively suggest that circRNA vaccines exert their potent antitumor effects by modulating the immune system to initiate a cascade of adaptive immune responses, ultimately eliminating tumor cells. Subsequently, we examined circRNA vaccine efficacy against infectious diseases. Findings demonstrated a significantly reduced viral load in mouse lungs post-vaccination, with sustained high antibody titers over extended periods The concurrent activation of innate and adaptive immune cells in infectious disease models further corroborated the broad and potent therapeutic efficacy of circRNA vaccines against both malignant tumors and infectious pathogens[16].

To further validate the potent anti-tumor and immune-activating effects of circRNA vaccines, we conducted in vitro verification experiments. To prepare high-purity circRNA and exclude non-specific immune interference, we combined RNase R-mediated digestion (to eliminate linear RNA precursors and fragments) with RNase III treatment (to specifically degrade dsRNA impurities), followed by secondary purification using a general silica column. Results demonstrated that purified circRNA successfully induced neoantigen expression in dendritic cells (DCs), while promoting DC activation and maturation, consistent with the findings of Huang et al.[43]. We further confirmed that circRNA-primed mature DCs could effectively present antigens to T cells and mediate the activation of antigen-specific T lymphocytes. These activated T cells subsequently secrete proinflammatory cytokines, including IL-1β, TNF-α, IL-6, and IFN-γ. ELISpot experiments validate that antigen-specific T cells can be generated, which in turn orchestrate a robust antitumor immune microenvironment and promote tumor cell killing, consistent with the results of Amaya et al.[15].

Collectively, our systematic meta-analysis and in vitro experimental validation consistently demonstrate that circRNA vaccines possess the capacity to elicit robust and sustained innate and adaptive immune responses, as well as to shape a proinflammatory tumor microenvironment. Endowed with multiple inherent advantages—including a covalently closed circular structure conferring superior stability, sustained and efficient neoantigen expression, and excellent in vivo tolerability—circRNA-encoded neoantigen vaccines represent a novel and promising therapeutic approach for the treatment of both malignant tumors and infectious diseases.

Notwithstanding these promising findings, several notable limitations and sources of substantial statistical heterogeneity must be acknowledged, which temper the interpretation and generalizability of our results. First, the meta-analysis revealed extreme statistical heterogeneity across the included studies for key immune and therapeutic outcome measures. Although we attempted to reduce heterogeneity through subgroup analysis, high heterogeneity persisted in some datasets. Comprehensive subgroup analysis indicated that this variability was primarily attributable to three interrelated factors: different routes of vaccine administration (e.g., intramuscular, subcutaneous, intravenous, intranasal), heterogeneity of disease models (e.g., different solid tumor types, viral pathogens), and diverse murine strains (C57BL/6, BALB/c, NOD/SCID) across the included studies. For instance, intranasal administration was associated with more pronounced local mucosal immune responses but relatively modest systemic immune effects, whereas intravenous injection elicited robust systemic immune activation.

Second, the current study is predominantly based on preclinical murine models, and the meta-analysis was restricted to animal experimental data due to the paucity of available clinical research on circRNA vaccines to date. There are inherent biological differences between murine models and humans, including disparities in immune system regulation (e.g., DC subsets, T cell receptor repertoires) and tumor microenvironment, which may result in inconsistent efficacy and safety profiles when translating circRNA vaccines from preclinical models to clinical applications. Although we supplemented our findings with in vitro validation using human PBMCs, this simplified system cannot fully recapitulate the complex in vivo microenvironment of the human body.

Finally, the detailed molecular mechanisms underlying the immune modulation of circRNA vaccines remain incompletely elucidated. While our results confirm that circRNA vaccines activate DCs and initiate adaptive immune responses, the specific molecular pathways by which LNP-circRNA is internalized by DCs and the potential crosstalk between circRNA and other innate immune sensors require further in-depth mechanistic investigation.

Despite these limitations, this study provides comprehensive and rigorous preclinical evidence for the therapeutic potential of LNP-delivered circRNA vaccines against both tumors and infectious diseases through systematic meta-analysis and in vitro validation with high-purity circRNA. Future research should focus on addressing the aforementioned limitations and heterogeneity to advance the development and clinical translation of circRNA vaccines. Priority areas include: conducting in-depth mechanistic studies to elucidate the molecular basis of circRNA uptake, cytoplasmic translation, and immune activation, thereby identifying potential molecular targets for enhancing vaccine efficacy; establishing more clinically relevant humanized mouse models to bridge the gap between preclinical and clinical research; and conducting large-scale, well-designed randomized controlled trials to validate the efficacy and safety of circRNA vaccines in human subjects. In conclusion, our study highlights the great potential of circRNA vaccines as a novel next-generation RNA vaccine platform, and with further optimization and in-depth validation, circRNA vaccines are expected to provide innovative and effective therapeutic options for human diseases.

Conclusion

This article demonstrates that LNP-delivered circRNA vaccines exhibit potent anti-tumor and anti-inflammatory effects, with minimal adverse reactions. Further experiments validate the capacity of circRNA to encode neoantigens, promote dendritic cell maturation, activate T cells, and stimulate the secretion of inflammatory cytokines, positioning it as a promising novel therapeutic approach for cancer treatment.

Data availability

The original contributions presented in the study are included in the article; further inquiries can be directed to the corresponding author.

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References

1. Liu, C. X. & Chen, L. L. Circular RNAs: characterization, cellular roles, and applications. *Cell* **185** (12), 2016–2034 (2022).
2. Memczak, S. et al. Circular RNAs are a large class of animal RNAs with regulatory potency . *Nature* **495** (7441), 333–338 (2013).
3. Ashwal-Fluss, R. et al. circRNA biogenesis competes with pre-mRNA splicing . *Mol. Cell.* **56** (1), 55–66 (2014).
4. Xia, P. et al. A circular RNA protects dormant hematopoietic stem cells from DNA sensor cGAS-mediated exhaustion. *Immunity* **48** (4), 688–701e7 (2018).
5. Hansen, T. B. et al. Natural RNA circles function as efficient microRNA sponges. *Nature* **495** (7441), 384–388 (2013).
6. Chen, C. K. et al. Structured elements drive extensive circular RNA translation. *Mol. Cell.* **81** (20), 4300–18e13 (2021).
7. Wesselhoeft, R. A., Kowalski, P. S. & Anderson, D. G. Engineering circular RNA for potent and stable translation in eukaryotic cells. *Nat. Commun.* **9** (1), 2629 (2018).
8. Yan, Y. et al. Non-viral vectors for RNA delivery. *J. Control Release*. **342**, 241–279 (2022).
9. Foster, J. B. et al. Purification of mRNA encoding chimeric antigen receptor is critical for generation of a robust T-cell response. *Hum. Gene Ther.* **30** (2), 168–178 (2019).
10. Yang, R. & Cui, J. Advances and applications of RNA vaccines in tumor treatment. *Mol. Cancer* **23**, 1 (2024).
11. Wesselhoeft, R. A. et al. RNA circularization diminishes immunogenicity and can extend translation duration in vivo. *Mol. Cell.* **74**(3), 508 -20.e4 (2019).
12. Chen, C. Y. & Sarnow, P. Initiation of protein synthesis by the eukaryotic translational apparatus on circular RNAs. *Science* **268** (5209), 415–417 (1995).
13. Sun, J. et al. Anti-tumour effect of neo-antigen-reactive T cells induced by RNA mutanome vaccine in mouse lung cancer. *J. Cancer Res. Clin. Oncol.* **147** (11), 3255–3268 (2021).
14. Xie, Z. et al. Cathepsin B in programmed cell death machinery: mechanisms of execution and regulatory pathways. *Cell. Death Dis.* **14** (4), 255 (2023).
15. Amaya, L. et al. Circular RNA vaccine induces potent T cell responses. *Proc. Natl. Acad. Sci. U S A*. **120** (20), e2302191120 (2023).
16. Qu, L. et al. Circular RNA vaccines against SARS-CoV-2 and emerging variants. *Cell* **185** (10), 1728–44e16 (2022).

17. Li, H. circRNA: a promising all-around star in the future. *Epigenomics* **15** (12), 677–685 (2023).
18. Ozpolat, B., Sood, A. K. & Lopez-Berestein, G. Liposomal siRNA nanocarriers for cancer therapy. *Adv. Drug Deliv Rev.* **66**, 110–116 (2014).
19. Moher, D. et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA Statement. *Open. Med.* **3** (3), e123–e130 (2009).
20. Hooijmans, C. R. et al. SYRCLE's risk of bias tool for animal studies. *BMC Med. Res. Methodol.* **14**, 43 (2014).
21. Alshehry, Y. et al. Investigation of the impact of lipid nanoparticle compositions on the delivery and T cell response of circRNA vaccine. *J. Controlled Release* **2025**, 381 (2025).
22. Wang, Y. et al. Synergically enhanced anti-tumor immunity of in vivo panCAR by circRNA vaccine boosting. *Cell. Rep. Med.* **6** (8), 102250 (2025).
23. Wang, F. et al. Circular RNA-based neoantigen vaccine for hepatocellular carcinoma immunotherapy. *MedComm* **5** (8), e667 (2020).
24. Li, H. et al. Circular RNA cancer vaccines drive immunity in hard-to-treat malignancies. *Theranostics* **12** (14), 6422–6436 (2022).
25. Huang, D. et al. Tumour circular RNAs elicit anti-tumour immunity by encoding cryptic peptides. *Nature* **625** (7995), 593–602 (2024).
26. Zhang, Y. et al. Small circular RNAs as vaccines for cancer immunotherapy. *Nat. Biomed. Eng.* **9** (2), 249–267 (2025).
27. Cai, Z. et al. CircRNA-loaded DC vaccine in combination with low-dose gemcitabine induced potent anti-tumor immunity in pancreatic cancer model. *Cancer Immunol. Immunother.* **74** (2), 68 (2025).
28. Li, H. et al. Intranasal prime-boost RNA vaccination elicits potent T cell response for lung cancer therapy. *Signal. Transduct. Target. Ther.* **10** (1), 101 (2025).
29. Jiang, Y. et al. GPC3-based circular RNA vaccine suppresses hepatocellular carcinoma progression by activating adaptive immune responses. *Hepatology* (2025).
30. Yue, X. et al. CircRNA based multivalent neuraminidase vaccine induces broad protection against influenza viruses in mice. *NPJ Vaccines*. **9** (1), 170 (2024).
31. Liu, X. et al. A single-dose circular RNA vaccine prevents Zika virus infection without enhancing dengue severity in mice. *Nat. Commun.* **15** (1), 8932 (2024).
32. Wu, J. et al. CircRNA vaccine encoding a chimeric immunogen of B6 and M1 demonstrates robust immune responses against MPXV. *Cell. Rep.* **44** (10), 116432 (2025).
33. Zhou, J. et al. Circular RNA vaccines against monkeypox virus provide potent protection against vaccinia virus infection in mice. *Mol. Ther.* **32** (6), 1779–1789 (2024).
34. Swingle, K. L. et al. Circular RNA lipid nanoparticle vaccine against SARS-CoV-2. *Proc. Natl. Acad. Sci. U S A.* **122** (38), e2505718122 (2025).
35. Zhang, W. et al. Bivalent circular RNA vaccines against porcine epidemic diarrhea virus and transmissible gastroenteritis virus. *Front. Immunol.* **16**, 1562865 (2025).
36. Zhang, Y. et al. Pulmonary delivery of small circular RNA vaccines for influenza prevention. *J. Control Release*. **387**, 114198 (2025).
37. Huang, K. et al. Circular mRNA vaccine against SARS-COV-2 variants enabled by degradable lipid nanoparticles. *ACS Appl. Mater. Interfaces*. **17** (3), 4699–4710 (2025).
38. Wan, J. et al. CXCL13 promotes broad immune responses induced by circular RNA vaccines. *Proc. Natl. Acad. Sci. U S A.* **121** (44), e2406434121 (2024).
39. Wan, J. et al. Circular RNA vaccines with long-term lymph node-targeting delivery stability after lyophilization induce potent and persistent immune responses. *mBio* **15** (1), e0177523 (2024).
40. Katrekar, D. et al. Efficient in vitro and in vivo RNA editing via recruitment of endogenous ADARs using circular guide RNAs. *Nat. Biotechnol.* **40** (6), 938–945 (2022).
41. Yi, Z. et al. Engineered circular ADAR-recruiting RNAs increase the efficiency and fidelity of RNA editing in vitro and in vivo. *Nat. Biotechnol.* **40** (6), 946–955 (2022).
42. Bowen, W. S. et al. Current challenges for cancer vaccine adjuvant development. *Expert Rev. Vaccines*. **17** (3), 207–215 (2018).
43. Huang, D. et al. Tumour circular RNAs elicit anti-tumour immunity by encoding cryptic peptides[J]. *Nature* **625** (7995), 593–602 (2023).

Author contributions

Ruohan Yang: wrote the main manuscript text and prepared figures. Lin Jia: Formal analysis, revise the manuscript. Jiuwei Cui: Conceptualization and investigation, final review, and editing of this manuscript. All authors reviewed the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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